

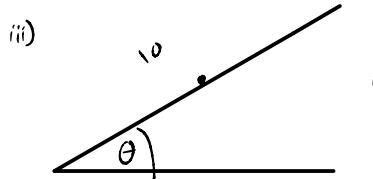
WEP 1

Consolidation

i) $P_e \rightarrow = 40 \text{ kW} = Fv \Rightarrow F = \frac{40000}{25} = 1600 \text{ N}$
 As $a = 0$ resistance to motion = 1600 N

ii) additional resistance = $800g \times \frac{1}{10} = 784.8 \text{ N}$

$\therefore$ creep out = 784.8×25
 $= 19.6 \text{ kW}$



$80g \text{ N} + 400 \text{ N} = F$
 $= 1684 \text{ N}$ Resistance

$\Delta KE = \frac{1}{2} \times 800 \times 25^2 = 250000$

$E_{res} = 1684d$ where $d = \text{distance travelled}$

$d = \frac{250000}{1684} \quad (E_{res} = \Delta KE)$

$= 146 \text{ m}$

$2 \Delta GPE_b = -1.5 \times 15g$
 $= -220.5 \text{ J}$

$\Delta GPE_a = 1.5 \times 15g$
 $= 58.8 \text{ J}$

$\Delta GPE_{\text{block}} = 1.5 \sin 40^\circ \times 5g$
 $= 47.24$

$\therefore \Delta GPE = (58.8 + 47.24 - 220.5)$
 $= -114.46 \text{ J}$ lost

ii) $\Delta GPE = -114$

$\Delta KE = \frac{1}{2} \times 24 \times v^2 = 12v^2$

$E_{res} = 1.5 \times (10 + 15) = 37.5 \text{ N}$

$\Delta KE = \Delta GPE - 37.5$
 $= 76.46$

$v^2 = \frac{76.46}{12}$

$v^2 = 6.413$

$v = 2.53 \text{ m s}^{-1}$

F

$$ii) KE_{\text{before}} = \frac{1}{2} \times 4 \times 2.53^2 + \frac{1}{2} \times 5 \times 2.53^2 = 28.80$$

$$GPE_{\text{initial}} = 0$$

$$KE_{\text{after}} = 0$$

$$GPE_{\text{final}} = 28.80 - 10J$$

$$28.80 - 10J = 5gd \sin 40^\circ + f_g d$$

$$28.80 = d(10 + 5g \sin 40^\circ + f_g)$$

$$d = 0.357m$$

$$3c) GPE = (6 + 5 \sin 40^\circ) \times 0.4 \times 9.8$$

$$= 36.1$$

$$GPE - KE = \text{Energy Lost}$$

$$= 2 \times 5 = 10J$$

$$KE = 26.1$$

$$= \frac{1}{2} \times 0.4 \times v^2$$

$$v = 11.4 m/s$$

$$ii) \text{vertically, } WJ = \Delta GPE = 12 \times 4.8 \times 6 = 705.6$$

$$\text{On roof, } \Delta E = (5 \sin 40^\circ \times 12g) + 50 \times 5$$

$$= 628J$$

$$\therefore \Sigma E = 1334J$$

$$\therefore P = \frac{E}{t} = 53.3 W$$

$$4i) 3760 = 240v$$

$$v = 1.75 m/s$$

$$ii) \text{a) } \Delta KE = \frac{1}{2} \times 380 \times (10^2 - 2^2)$$

$$= 18240$$

$$\text{b) } \Delta GPE = 380 \times 9.8 \times (20 - 5)$$

$$= 55860J \text{ lost}$$

$$iii) GPE - KE = \text{work done via } F_{\text{res}}$$

$$= 37620$$

$$37620 = Fd$$

$$= (150N + F_{\text{nonbraking}}) \times 180$$

$$E_{\text{nonbraking}} = \frac{37620 - 15000}{180} = 22620$$

$$iv) F_{Re} = 376.2$$

$$KE_{\text{before}} = \frac{1}{2} \times 380 \times 10^2 = 19000 \text{ J} \quad KE_{\text{after}} = 0$$

$$GPE_{\text{before}} = 0 \quad GPE_{\text{after}} = 380gh$$

$$380gh + 19000 = \frac{376.2 h}{\sin 20^\circ}$$

$$h \left(380g + \frac{376.2}{\sin 20^\circ} \right) = 19000$$

$$h = 3.93 \text{ m}$$

Exam Questions

$$1) 360 \text{ kg min}^{-1} = 6 \text{ kg s}^{-1}$$

$$\frac{KE}{\text{sec}} = \frac{1}{2} \times 6 \times 6.2^2 = 116.1 \text{ J}$$

$$\frac{\Delta GPE}{\text{second}} = 6 \times g \times 4.5 = 270.5$$

$$\therefore P = \frac{270.5 + 116.1}{1} = 386.6 \text{ W}$$

$$2a) KE = \frac{1}{2} mv^2 \\ 16000 = \frac{mv^2}{2} \\ mv^2 = 32000 \\ v = \sqrt{\frac{32000}{m}} \\ = 16 \text{ ms}^{-1}$$

$$b) 150 + 800 \times \frac{1}{20} = 542 \text{ N} \\ \text{so } F_{\text{net}} = 542 \text{ N}$$

$$P = Fv \\ 15000 = 542v \\ v = 28 \text{ ms}^{-1}$$

$$c) 18000 = 30 \times F \\ F = 600 \text{ N}$$

$$a = 0$$

$$\therefore R + 3R = 600 \\ R = 150 \text{ N}$$

$$\text{car's resistance to motion} = 375 \text{ N}$$

$$T = 375 \quad 800$$


$$125 \quad T$$


$$T = 125 \text{ N}$$